



SMIC 0.18um Mixed Signal Enhanced Calibre LVS Release Note

Version: **V1.11_1**

Version Date: **2018-09-13**

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1.0 Scope

This document describes the Calibre LVS of 0.18um Mixed Signal 1P6M (1P5M,1P4M,1P3M) with MIM Salicide 1.8V/3.3V release. Specific updates to Calibre LVS release are discussed within.

1.1 Contents

LVS deck:

SMIC_CalLVS_018MSE_1833_V1.11_1.lvs

Empty subckt netlist : empty_subckt.sp

1.2 Calibre LVS deck options setup

```
//-----  
// *OPTION 1: Define TOP Metal. The value can be 6, 5, 4, 3  
#DEFINE TOPMETAL 6  
//-----  
// *OPTION 2: Define ERC Check or Not. The value can be TRUE or FALSE(Upper Case).  
#DEFINE ERCCHECK TRUE  
//-----  
// *OPTION 3: Define Top Metal type. The value can be 9K, 22K, 34K or 40K.  
// * 9K(Upper Case) : Sheet Resistance of Top Metal is 0.0279 ohm/sq.  
// * 22K(Upper Case): Sheet Resistance of Top Metal is 0.015 ohm/sq.  
// * 34K(Upper Case): Sheet Resistance of Top Metal is 0.01 ohm/sq.  
// * 40K(Upper Case): Sheet Resistance of Top Metal is 0.0083 ohm/sq.  
#DEFINE TM_TYPE 9K  
//-----  
// *OPTION 4: Define Model name of the Memory for Virage or Artisan Library.  
// *TRUE(Upper Case): Model name of sram device is N18, P18.  
// *FALSE(Upper Case): Model name of sram device is NPG, NPD, PL.  
#DEFINE VIR_ART FALSE // *Can be TRUE or FALSE  
//-----  
// *OPTION 5: Define MIM type. The value can be 1, 2.  
  
// *1: Model name is mim or mim_ckt, The C0 is 0.969 fF/um2.
```

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```
/**2: Model name is mim2 or mim2_ckt, The C0 is 1.99 fF/um^2.
#DEFINE MIM_TYPE 1
//-----
/**OPTION 6: Define RC run or not; The value can be TRUE or FALSE.
#DEFINE RC_RUN FALSE
//-----
/**OPTION 7: Extract EFUSE_res or not;
/**Default value is FALSE.
#DEFINE EFUSE_RES TRUE
//-----
/**OPTION 8: Define BJT_MIS_MODE; The value can be 0 or 1.
#DEFINE BJT_MIS_MODE 1
//-----
/**OPTION 9: Define MIM_MIS_MODE; The value can be 0 or 1.
#DEFINE MIM_mismod 1
//-----
/**OPTION 10: Define RES_MIS_MODE; The value can be 0 or 1.
#DEFINE RES_mismod 1
//-----
/**OPTION 11: Define VAR_MIS_MODE; The value can be 0 or 1.
#DEFINE VAR_MIS_MODE 1
//-----
/**OPTION 12: Compare diode number and connectivity or not for Parasitic DIODE.
//The value can be TRUE or FALSE(Upper Case).
/**Default value is FALSE.
#DEFINE PARASI_DIO_CMP_DNW FALSE
//-----
/**OPTION 13: Compare diode number and connectivity or not for Parasitic DIODE.
//The value can be TRUE or FALSE(Upper Case).
/**Default value is FALSE.
#DEFINE PARASI_DIO_CMP_NW FALSE
//-----
/**OPTION 14: Compare parameters for parasitic DIODE.The value can be TRUE or FALSE(Upper Case).
/**Default value is FALSE.
#DEFINE PARASI_DIO_Property_CMP_DNW FALSE
//-----
/**OPTION 15: Compare parameters for parasitic DIODE.The value can be TRUE or FALSE(Upper Case).
/**Default value is FALSE.
#DEFINE PARASI_DIO_Property_CMP_NW FALSE
//-----
```

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```
/*OPTION 16: Turn on nwell floating check or not. The value can be TRUE or FALSE(Upper Case).
/*Nwell resistor and BJT are not checked.
/*Default value is FALSE.
#define NWELL_FLOATING_CHECK FALSE
//-----
/*OPTION 17: Turn on nwell not connecting to power check or not. The value can be TRUE or FALSE(Upper Case).
/*Nwell resistor and BJT are not checked.
/*Default value is FALSE.
#define NWELL_NOT_TO_POWER_CHECK TRUE
//-----
/*OPTION 18: Turn on pwell not connecting to ground check or not. The value can be TRUE or FALSE(Upper Case).
/*Default value is FALSE.
#define PWELL_NOT_TO_GROUND_CHECK TRUE
//-----
/*OPTION 19: Define whether to use CASE-SENSIVITY.The value can be TRUE or FALSE(Upper Case).
#define CASE_SENSITIVITY FALSE
//-----
/*OPTION 20: Define the back annotation flow.The value can be 1 or 2,Default value is 1.
/*1: extrview flow,LVS will reserve some extra parameters for back annotation.
/*2: netlist flow,LVS will not reserve the back annotation parameters.
#define Back_Annotation_Flow 2
//-----
/*OPTION 21: compare PJ parameter or not for DIODE.The value can be TRUE or FALSE(Upper Case).
/*Default value is FALSE.
#define DIO_PJ FALSE
//-----
```

1.3 LVS Mark Layer drawn

Notice: Customer could get the information about LVS mark layer of any devices from the head of LVS runset.

For example: Resistor

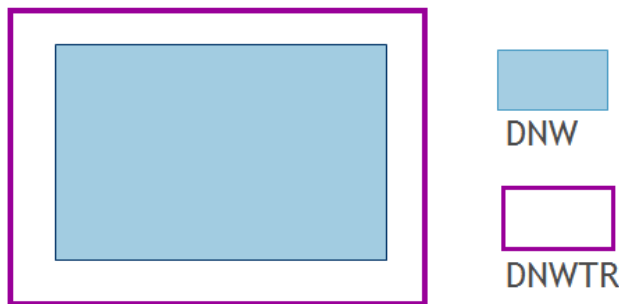
```
// Resistor Type: //
// //
// rnwsti          Nwell under STI          (RESNW*NW-RESCKT) //
// rnwaa           Nwell under AA           (RESNW*NW*AA*SAB-RESCKT) //
// rndifsab        Non-silicide N+ Diffusion (RESAA*SN*AA*SAB-RESCKT) //
```

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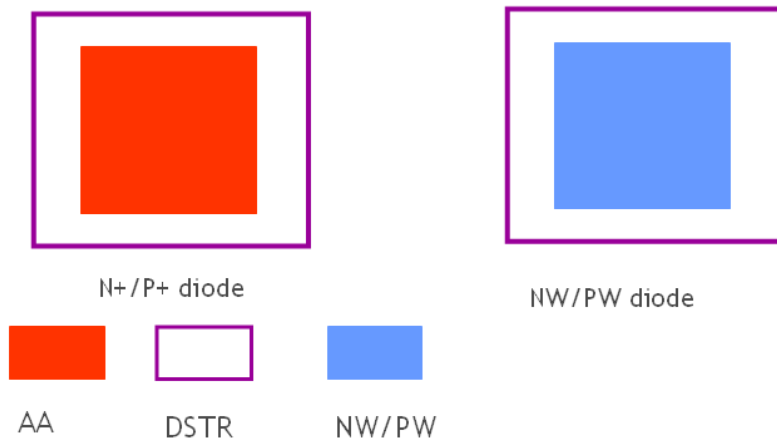
DNWTR(19;2)

- Purpose: LVS mark layer for DNW MOS extraction
- Usage:
 - 1) DNWTR must enclose DNW
 - 2) DNWTR must not cover other devices except dnwdio and dnw MOSFET



DSTR(138;0)

- Purpose: LVS mark layer for diode extraction
- Usage:
 - 1) DSTR should enclose the diffusion region for diffusion diode
 - 2) DSTR should enclose the well for well diode

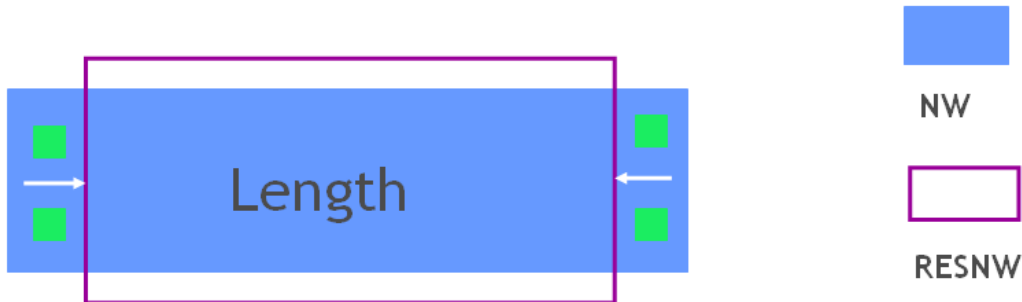


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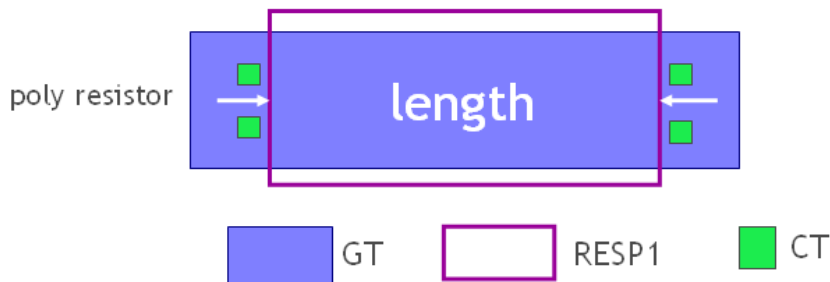
RESNW(95;0)

- Purpose: LVS mark layer for NW resistor extraction
- Usage: 1) RESNW should only cover the NW resistor body region
- 2) RESNW should not cover other device



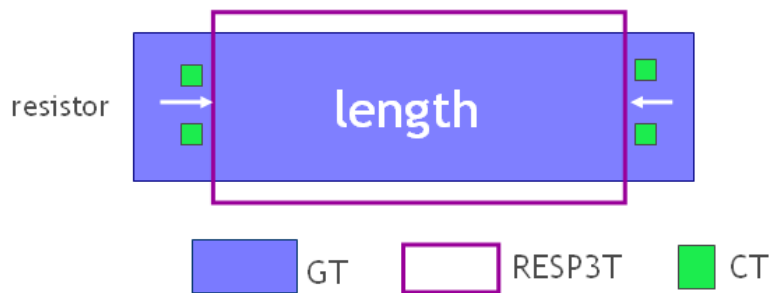
RESP1(96;0)

- Purpose: LVS mark layer for poly resistor extraction
- Usage: 1) RESP1 must only cover the poly resistor body region
- 2) RESP1 must not cover other devices



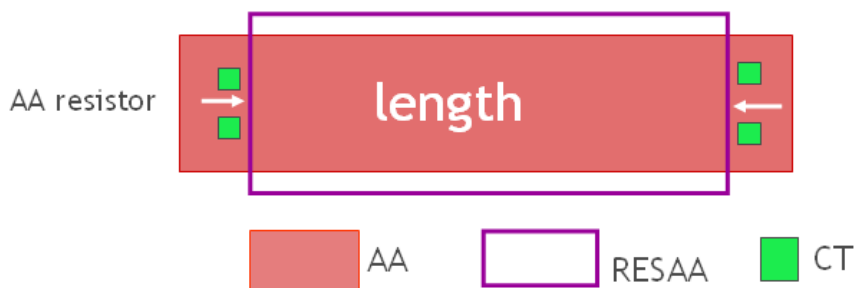
RESP3T(96;1)

- Purpose: LVS mark layer for 3T resistor extraction
- Usage: 1)RESP3T must at least cover the resistor body region
2)RESP3T must not cover other devices



RESAA(97;0)

- Purpose: LVS mark layer for AA resistor extraction
- Usage: 1)RESAA must only cover the AA resistor body region
2)RESAA must not cover other devices



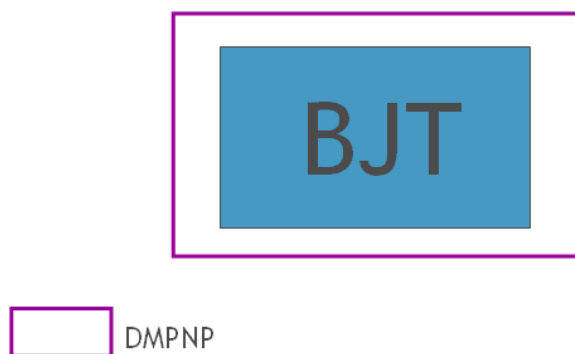
M1R~MnR

- Purpose: LVS mark layer for metal resistor extraction
- Usage: 1) MnR should only cover the metal resistor body region
- 2) MnR should not cover other device



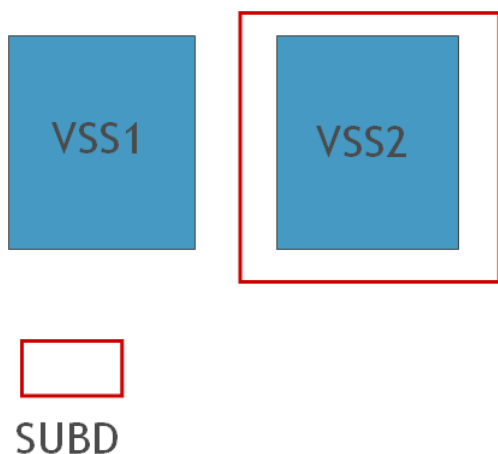
DMPNP(134;0)

- Purpose: LVS mark layer for BJT extraction
- Usage: 1) DMPNP should at least cover the whole BJT region and BJT collector terminal
- 2) DMPNP should not cover other device



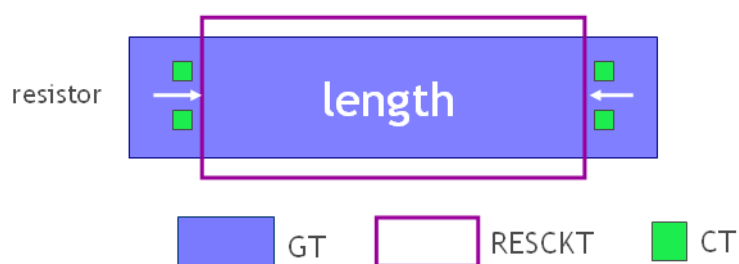
SUBD(131;1)

- Purpose: LVS mark layer for different substrate
- Usage: 1) SUBD should be drawn when there is more than one substrate ground



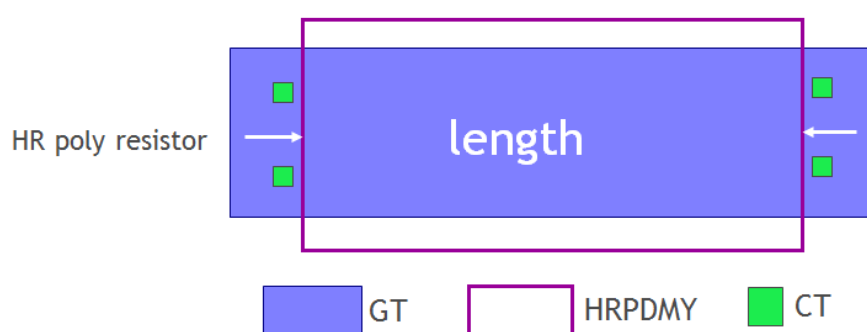
RESCKT(131;3)

- Purpose: LVS mark layer to distinguish R model resistor and SUBCKT resistor
- Usage: 1) RESCKT must at least cover the resistor body region
2) RESCKT must not cover other devices



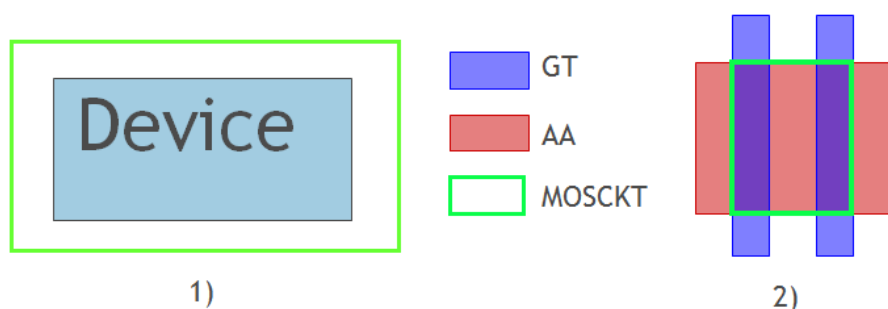
HRPDY(210;0)

- Purpose: LVS mark layer for HRP Resistor extraction
- Usage: 1)HRPDY must at least cover the HRP resistor body region
2)HRPDY must not cover other devices



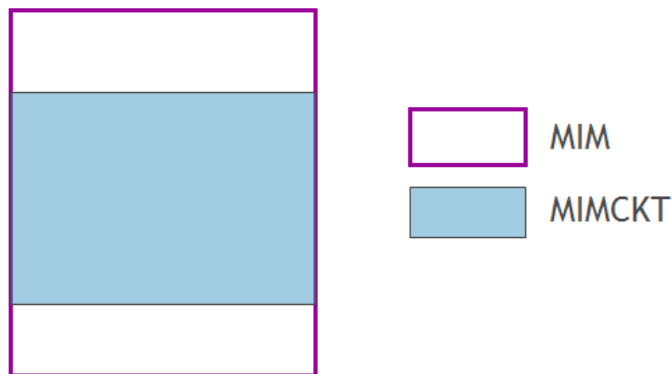
MOSCKT(131;2)

- Purpose: LVS mark layer for mismatch devices extraction
- Usage:
 - 1)for mismatch devices, MOSCKT must at least cover the whole device region
 - 2)MOSCKT must not cover other devices



MIMCKT(131;6)

- Purpose: LVS mark layer for mim capacitor width calculation
- Usage: 1)MIMCKT must partially cover the MIM layer region
2)MIMCKT must not cover other devices



1.4 Tolerance comparison

Compare Tolerance and Device Combine

The default tolerance used for distance or dimension is 5%.

The default tolerance used for integer parameter is 0%.

MOS:

Parallel combined allowed

Series combined not allowed

LVS Compare: w&L

BJT:

Parallel combined allowed

Series combined not allowed

LVS Compare: model name

RES:(subckt model)

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Parallel combined allowed

Series combined allowed

LVS Compare: W,L

RES:(R model)

Parallel combined allowed

Series combined allowed

LVS Compare: R

DIODE

Parallel combined allowed

Series combined not allowed

LVS Compare: area

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1.5 QA Details

Test Case Type	Results	Known Problems
Test Pattern	pass	NA
SMIC PDK	pass	NA

1.6 FAQ

@ Calibre LVS & DRC licenses are needed.

@ Calibre Version for test: V2013.4

@ Please include “empty_subckt.sp” file in the schematic netlist when run the subckt devices.

Syntax for how to include this file:

```
.include ./empty_subckt.sp
```

1.7 Revision History

```
// 1.11_1 2018/09/12 Liu Li based on the following Process Documents: //
//          DESIGN RULE:      TD-L018-DR-2001 Rev.:12 //
//          (Tech Dev. Rev.:1.19_REV1) //
//          SPICE MODEL:      TD-MM18-SP-2004 Rev.:2 //
//          (Tech Dev. Rev.:1.11_0) //
//          What's modified: //
//          1.follow model update, add npd33_ckt,npg33_ckt and //
//          pl33_ckt device //
//          2.follow DRM's device true table update,update none //
//          layer definition for diobpw,dnwdio,n18,n33,nmvt18 //
//          and nmvt33 device //
//          .. //
```

What's modified:

1. follow model update, add npd33_ckt,npg33_ckt and pl33_ckt device
2. follow DRM's device true table update, update none layer definition for diobpw, dnwdio,n18,n33,nmvt18 and nmvt33 device

1.8 Important Notes on usage and Known problems

- 1) The default tolerance used for distance or dimension is 5%.

The default tolerance used for integer parameter is 0%.

The default tolerance can be changed according to the design requirement.

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- 2) Please set LVS with option "XRC_RUN"=FALSE first to guarantee the LVS result clean or not. If LVS result is clean with "XRC_RUN"=FALSE, then set the option "XRC_RUN"=TRUE to run LVS+RC extraction for post-layout simulation.
- 3) The layer SUBD(131;1) is used to logically separate the substrate for LVS. USE of SUBD doesn't in any way affect the mask build process or otherwise provide any type of real physical isolation. Be aware of the possible problems of using this layer.
- 4) Below conventional ERC rules are included in LVS deck. Once there is any violation of ERC related, designer should review it and then decide to waive it or not according to design requirement.
 - A) Report nodes with a path to power but not ground
 - B) Report nodes with a path to ground but not power
 - C) Report nodes without a path to both power and ground
 - D) Report nodes without a path to any labeled net
 - E) Select pwell pick-up layer ptd connect to power nets.
 - F) Select nwell pick-up layer ntd connect to ground nets.
 - G) NMOS S/D connect to both Power and Ground nets
 - H) PMOS S/D connect to both Power and Ground Nets.
 - I) Report pwell pick-up layer ptap_outside_dnw connect to non-ground nets.
 - J) Report floating nwell with no pick up
 - K) Report nwell(exclude BJT and NW resistor) not connecting to power net

L) Report pwell not connecting to ground net

5) Please enter the LVS POWER and GROUND name in below corresponding place of LVS deck to get right LVS/ERC results.

LVS GROUND NAME "VSS" "SAVSS?" "?GND?" "?VSS?" "?vss?" "?gnd?"

LVS POWER NAME "VDD" "SAVDD?" "?VDD?" "?VCC?" "?vcc?" "?vdd?"

6) Text layer and metal layer are used for LVS text and port recognition.

Text Layer	Metal Layer
M1TXT(141;0)	M1(61;0)
M2TXT(142;0)	M2(62;0)
M3TXT(143;0)	M3(63;0)

